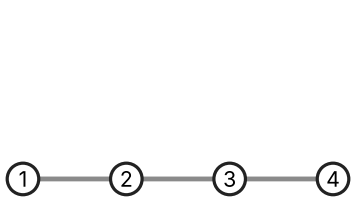
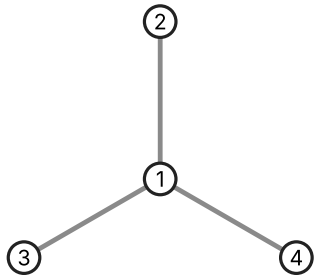
path



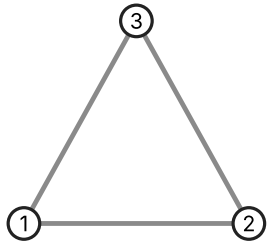
N=1: faithful
N=2: faithful

star



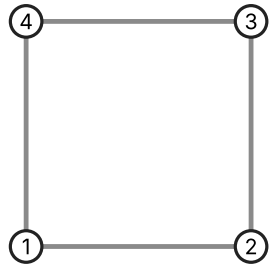
N=1: faithful
N=2: faithful

triangle



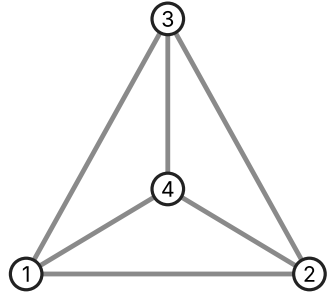
N=1: relation
N=2: faithful

square



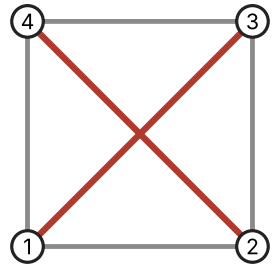
N=1: faithful
N=2: faithful

K4, planar



N=1: relation
N=2: faithful

K4, crossed



not a homomorphism

Red: independent edges whose segments cross (their twists fail to commute). Faithfulness: on all words of length <= 4 or 5.